

GEOOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Ottertail Power Company 215 S. Cascade St Fergus Falls, MN 56537 ATTN: Bill Gronwold	2025-12-17	QU-0221
		Reference
		US MN Excel Plastics

Understanding of Project

OTPCO is supporting a commercial customer located at 1021 International Drive in Fergus Falls, MN, in evaluating options for upgrading their facility’s heating and cooling systems. The existing building includes approximately 30,000 square feet of production space, 34,000 square feet of warehouse area, and 7,000 square feet of office space. The facility currently utilizes an electric cable thermal storage system embedded in the concrete floor, which is experiencing performance issues. The customer is currently served under OTPCO’s fixed time-of-delivery rate (10:00 PM – 6:00 AM).

At present, air conditioning is limited to the office area, with no ductwork serving the remainder of the facility. The customer is exploring the potential implementation of a geothermal heat pump system to replace or supplement the existing heating system. To support this effort, OTPCO has requested a Geo Optimize feasibility study to evaluate the technical and economic viability of a geothermal solution, including anticipated energy efficiency gains, operational cost impacts, and long-term sustainability benefits for the facility.

Scope of Work

Feasibility

- Develop a detailed hourly energy model of the proposed building.
 - The model may include several iterations of the proposed building to determine the ventilation strategy, glass specifications and other potential energy efficiency measures that will result in the most cost-effective ground heat exchanger for the facility.
- Ground Loop Design (GLD) model to size the ground heat exchanger (GHX)
- Review of the geology of the site to determine the thermal properties of the soil and the most cost-effective drilling depth for potential drilling contractors in the area.
- Provide size and cost estimates for either a horizontal and vertical GHX options to support a life-cycle cost analysis (LCCA), including operational costs and a financial evaluation of ROI, NPV, and simple payback.

Information Required

- Detailed construction drawings and specifications for the proposed building, including:
 - Insulation values of walls, floor, and roof.
 - Window specifications (solar heat gain coefficient (SHGC) and U-value).
 - Estimated occupancy schedule of the building.
 - Proposed mechanical system for the building, including ventilation requirements and specifications.
 - Electrical and lighting drawings and specifications for the building.
- Details pertaining to the production process, including:
 - Hours of operation,
 - Number of people on production floor,
 - Electrical load and/ or heat output of process loads,
- Site plan indicating land area available for construction of the GHX.
- Geotechnical report.

Deliverables

- Preliminary GHX options and size/cost for feasibility analysis in PowerPoint format.

Pricing

Description	Rate/hr	Qty	Line Total
Energy Model	\$150.00	48	\$7,200.00
GLD Model & Review of Geology	\$175.00	8	\$1,400.00
Financial Analysis	\$175.00	12	\$2,100.00
Presentations & Meetings	\$175.00	12	\$2,100.00
Total (USD)			\$12,800.00

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